

$$f(x) = \left. \begin{array}{ll} \sin 2x, & x \geq 0 \\ a + bx, & x \leq 0 \end{array} \right\}$$

$$f'(x) = \left. \begin{array}{ll} 2\cos 2x, & x \geq 0 \\ b, & x \leq 0 \end{array} \right\} \quad \begin{array}{l} \rightarrow x=0^- \Rightarrow f'(x) = 2\cos(2 \cdot 0^-) \\ \quad \quad \quad = 2 \\ \rightarrow x=0^+ \Rightarrow f'(x) = b \end{array}$$

$$\boxed{b=2}$$

Since it's diff, it's continuous

$$\rightarrow \sin 2x = a + bx \quad \text{at } x=0$$

$$\rightarrow \cancel{0 = a(0) + b} \Rightarrow 0 = a + b(0) \\ \Rightarrow \boxed{a=0}$$

$$\boxed{a+b=2}$$